Shafia Ameeruddin

Email | LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

- AI Engineer specializing in Generative AI, agentic AI systems, and full-stack product engineering, with hands-on experience designing and deploying production-ready AI applications.
- Built multimodal Retrieval Augmented Generation (RAG) platforms, autonomous multi-agent workflows, and evaluation-driven AI systems using Python, FastAPI, React, LangGraph, and modern LLM orchestration.
- Combines strong software engineering fundamentals with end-to-end product ownership to deliver scalable, reliable AI solutions from architecture through deployment.

EDUCATION

Jamia Hamdard — **New Delhi, India**
Bachelor of Technology in Computer Science and Engineering | CGPA: 8.55/10 — Sep 2022 - Jun 2026
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Object-Oriented Programming, DBMS

TECHNICAL SKILLS

- **Languages**: Python, JavaScript, SQL
- **AI/ML**: Large Language Models (LLMs), Agentic AI, Generative AI, Prompt Engineering, NLP, Computer Vision, RAG, LangChain, LangGraph, Hugging Face Transformers, PyTorch, Scikit-learn
- **Software Engineering**: FastAPI, React, Node.js, REST API, Streamlit
- **Data & Databases**: NumPy, Pandas, Matplotlib, Plotly, MySQL, SQLite, MongoDB
- **Cloud, DevOps & Tools**: AWS, Docker, Git, GitHub, CI/CD

WORK EXPERIENCE

CyArt — **Remote**
AI Engineer Intern — Sep 2025 - Dec 2025

- Engineered a concurrent Python news analytics pipeline processing 494+ headlines from 20 RSS sources, improving ingestion performance by 20%, achieving 19 headlines/sec throughput, and generating 1,877+ sentiment classifications via 3 NLP engines.
- Implemented a real-time OpenCV surveillance system monitoring 4 concurrent camera streams at ~7.4 FPS, using MOG2 motion detection and 2 image-integrity checks to flag potential camera compromise.
- Researched and benchmarked XTTS-v2 and VALL-E-X (text-to-speech models) through 15+ voice-cloning experiments, building automated evaluation pipelines that achieved 0.965 speaker similarity while assessing latency, stability, and speech quality.

Edunet Foundation — **Remote**
Front-End Web Developer Intern — Aug 2025 - Sep 2025

- Developed and deployed '**CineScope**', a full-stack movie discovery platform using HTML, CSS, JavaScript, Node.js, & Express, integrating the OMDb API to deliver 8 user-facing features including search, reviews, ratings, and personalized watchlists.
- Engineered an Express proxy keeping the OMDb API key server-side, implementing 6 client-side validation and verification checks, 3 error-recovery mechanisms, and resolving 5 production-critical issues, improving reliability and deployment readiness.

PROJECTS

Learnify AI | FastAPI • React • LangChain • MongoDB • JWT — GitHub | Live | Demo

- Architected a production-grade multimodal AI tutoring platform ingesting 3 file formats (PDFs, PPTs, and TXTs) into a user-scoped FAISS RAG pipeline with 3 level-adaptive prompts, source-page citations, voice I/O, and responses in 6 languages.
- Orchestrated 3 hot-swappable LLMs (Gemini, Groq, Ollama) with real-time DeepFace emotion detection, mapping 7 emotions to 4 learning states, alongside a 5-tier adaptive quiz engine in React and 6 AI-powered, document-driven mini-games.

Stratix | FastAPI • Streamlit • LangGraph • LangSmith • SQLite — GitHub | Live | Demo

- Built an autonomous SEO market intelligence platform that turns seed keywords into strategy reports using 5 research tools, a 9-node stateful LangGraph pipeline with a ReAct research agent, and 2 human approval checkpoints.
- Established quality controls to flag unreliable findings: deterministic confidence scoring, a quality gate, an LLM critic, bounded retries, 3 LLM-as-judge evaluations and SQLite checkpointing, validated by 238 automated tests.

MediScan | PyTorch • torchvision • Streamlit • PyMuPDF • BeautifulSoup4 — GitHub | Live | Demo

- Developed a medical vs. non-medical image classifier by fine-tuning ResNet18 in PyTorch with staged transfer learning, class-balanced sampling, and early stopping, achieving 100% accuracy (101/101) on a validation set.
- Implemented Streamlit and CLI inference interfaces supporting image, PDF, and URL inputs with ~52 ms CPU latency (TTA disabled) and a 0.60 confidence threshold flagging uncertain predictions.

CartVerse | Hugging Face Transformers • TextBlob • Streamlit • MySQL • Plotly — GitHub | Live | Demo

- Built a Walmart shopping assistant with 2 independent pipelines (MoodCart + AutoCart), mapping 26 emotional states to product categories and generating frequency-based carts across 10 mock user purchase histories.
- Engineered a 3-tier sentiment fallback (regex, DistilBERT, TextBlob) and a SerpAPI Walmart search layer with a 3600-second cache and 429 retry handling, keeping the app functional under constraints.

PacketWatch | Scikit-learn • Scapy • NumPy • Pandas • Streamlit — GitHub | Demo

- Engineered a Scapy-based feature extraction pipeline producing 27 statistical flow features from live Wi-Fi captures, enabling threat classification within a 20-second capture window.
- Trained a 100-estimator Random Forest classifier with serialized imputer and scaler, delivering sub-second inference on 100-row CSV network logs through a Streamlit dashboard.

LEADERSHIP & EXTRACURRICULAR EXPERIENCE

Google Developer Groups on Campus, Jamia Hamdard — **New Delhi, India**
Social Media Manager & Team Lead — Sep 2024 - Aug 2025
- Led a 10-member social media team, revamping the organization's social media presence and executing data-driven content strategies, driving 65% growth in digital reach, 40% higher content efficiency, and 5+ new industry partnerships.

PR & Social Media Associate — Jan 2024 - Sep 2024
- Increased social media engagement by 50% through creative content creation and targeted campaigns; promoted 15+ annual workshops and events, strengthening community participation and visibility.

IEEE Jamia Hamdard Student Branch — **New Delhi, India**
Student Coordinator — Aug 2023 - Jul 2025
- Contributed to 25+ flagship IEEE events impacting 700+ participants, boosting engagement and branch visibility by 40% and event participation by 30% through design support, hosting, and cross-functional coordination.

The Department of Placement and Alumni Affairs, Jamia Hamdard — **New Delhi, India**
Student Coordinator — Mar 2024 - May 2025
- Facilitated industry connections for 200+ students and supported campus recruitment coordination with Placement Cell leadership.

ICIDSSD '25, Jamia Hamdard — **New Delhi, India**
Student Volunteer — Apr 2025
- Coordinated conference operations and hosted one track at the 5th International Conference on ICT for Digital, Smart and Sustainable Development (ICIDSSD '25), supporting session execution, scripts, and award arrangements with a 50-member team at an 8-track, 2-day conference with 100+ participants.

DSSYWLC-24, IEEE Delhi Section SAC — **New Delhi, India**
Participant — Feb 2025
- Participated in the Delhi Section Students' Young Professionals, Women in Engineering, Life Members' Congress (DSSYWLC-24), attending technical sessions, workshops, and interactive activities focused on technology and professional development.

Enord — **New Delhi, India**
Product Launch & Event Assistant — Jan 2024
- Coordinated pre-event logistics for DRONE DAY, an aerospace product launch serving 150+ participants, and analyzed attendee feedback to inform future product launches and marketing strategies.

CERTIFICATIONS

- **IBM** SkillsBuild Artificial Intelligence Fundamentals | Credentials
- **NVIDIA** AI for All: From Basics to GenAI Practice | Credentials
- **IBM** SkillsBuild Web Development Fundamentals | Credentials
- **IBM** SkillsBuild Cybersecurity Fundamentals | Credentials
- **LetsUpgrade** Python Bootcamp | Credentials
- **LetsUpgrade** SQL Bootcamp | Credentials

HACKATHONS & COMPETITIONS

- **Walmart** Sparkathon 2025 Participant | Credentials
- **Adobe** India Hackathon 2025 | Credentials
- **Google** Solution Challenge 2025 | Credentials